

Critical Questions Generation

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Abstract

Large language models (LLMs) now produce fluent prose, yet they still struggle to interrogate that prose with *critical questions*. The ACL 2025 Critical Question Generation (CQG) shared task asks systems to pose three concise questions that could undermine a given argument. We cast CQG as a reinforcement-learning (RL) problem and adapt **Group-Relative Policy Optimisation** (GRPO) with rank-64 LoRA adapters. A composite reward encourages (i) strict XML formatting, (ii) exactly three questions, and (iii) semantic *usefulness* judged by an auxiliary LLM. Fine-tuning lifts a 3-billion-parameter Qwen model from the bottom of the leaderboard to mid-table ($+0.13 F_{LLM}$), and pulls an 8-billion-parameter Llama within 0.0 pp of the 405 B frontier—after only three RL epochs. These results showcase the *head-room* of preference-based RL for CQG and highlight the need for richer, rubric-driven evaluation protocols to cope with the task’s inherently subjective nature. Code and prompts: https://github.com/Myriam2002/Critical_Question_generation

1 Introduction

Critical thinking is ignited by questions, not answers. When readers confront an argumentative passage, they mentally fire off *critical questions* (CQs) such as:

“Could the correlation be spurious?” “Does the author overlook alternative causes?”

Such queries expose hidden assumptions and compel the authors to defend weak points. Calvo Figueras and Agerri (2024) formalised **Critical Question Generation** (CQG) and released an initial dataset derived from Walton’s argumentation schemes (Walton et al., 2008). Yet CQG remains under-explored: most question-generation work targets factual recall or reading comprehension rather than adversarial scrutiny.

The ACL 2025 shared task pushes CQG forward with 186 argumentative passages and 4,136 annotated questions, each labelled USEFUL, UNHELPFUL, or INVALID. Systems must generate three questions per passage and are scored by the proportion judged USEFUL. Two challenges stand out:

1. **Subjective reward.** Usefulness is a matter of reader knowledge and scepticism; reference coverage is sparse.
2. **Structured output.** The organiser requires exactly three questions.

We tackle both challenges with a *single* approach: **reinforcement learning**. Our contributions are:

- **Composite-reward GRPO.** We adapt Group-Relative Policy Optimisation (GRPO) to CQG, combining formatting, structure and semantic-usefulness signals.
- **Scalable LoRA tuning.** Rank-64 adapters are injected into seven projection layers, yielding memory-efficient fine-tuning for 3–8 B backbones.
- **Empirical gains and analysis.** RL lifts small models substantially and closes most of the gap to a 405 B baseline; reward-component curves reveal where progress stalls.

Our findings suggest that preference-based RL can drive CQG performance higher, but a richer human-in-the-loop evaluation—beyond similarity—will be critical for future advances.

2 Background

2.1 Critical Question Generation (CQG)

Calvo Figueras and Agerri (2024) introduced CQG as the task of producing pointed questions that expose hidden premises or missing evidence in an

argument. Their pilot corpus leverages Walton’s argumentation schemes and associated critical questions (Walton et al., 2008), but coverage remains sparse and evaluation noisy. The ACL 2025 shared-task expands this setting to 186 passages with 4 136 human-authored CQs and a three-way usefulness label.

2.2 Question Generation Beyond Fact Recall

Early question-generation (QG) work relied on rule-based transformations (Heilman and Smith, 2010), while neural seq2seq models later improved fluency and diversity (Du et al., 2017). Datasets such as INQUISITIVE (Ko et al., 2020) have shifted attention towards *why/how* questions that test discourse-level understanding. Nevertheless, these systems are optimised for information retrieval or reading-comprehension, not for adversarially probing an argument’s strength—precisely the gap CQG aims to fill.

2.3 Reinforcement Learning for Text Generation

Preference-based reinforcement learning has become the de-facto standard for aligning language-model outputs with human. Starting with RLHF using PPO (Ouyang et al., 2022), newer methods such as DPO (Rafailov et al., 2023) and RLAIIF (Bai et al., 2022) replace costly human labelling with LLM or rule-based feedback. Most recently, *Group-Relative Policy Optimisation* (GRPO) (Shao et al., 2024) stabilises training via group-wise normalisation of rewards. Parameter-efficient adapters (e.g. LoRA (Hu et al., 2023)) make such RL tuning feasible on commodity hardware. Our work is the first to apply GRPO+LoRA to Critical Question Generation, combining hard-constraint rewards (XML format, question count) with a semantic usefulness signal supplied by similarity measure and an auxiliary LLM.

3 Methodology

3.1 Data

Each critical question (CQ) is labeled as one of three categories: **USEFUL**, **UNHELPFUL**, or **INVALID**.

- **USEFUL**: Questions that directly challenge or probe the argument.
- **UNHELPFUL**: Valid questions that do not meaningfully challenge the argument.

- **INVALID**: Irrelevant or incorrect questions.

All examples have been annotated by humans according to publicly available guidelines (see Appendix A).

Below is an example argumentative intervention, followed by one question of each type:

Argumentative Text

MT: “We have indeed required those products to have very large warnings on them, telling people about their salt and fat content. And we don’t allow misleading claims—fatty products can’t be marketed as ‘healthy.’ I’ve got a letter here from a credit card company with two figures that jump out: 0% per annum on new balance transfers, 0% per annum on new money transfers. But in the fine print, you’ll see hefty fees—and very quickly you’ll end up paying a large charge. Do you think it’s responsible to encourage people to misread a deal like that? I have all the details right here.”

USEFUL question: What is the basis for comparing the credit card company’s marketing to food-warning labels, and is that comparison valid?

UNHELPFUL question: How does the speaker know the credit card advertisement is irresponsible, and what evidence supports this claim? (*The answer is already stated in the text.*)

INVALID question: Is the speaker’s argument logically sound, or are there any gaps or flaws? (*This question could apply to any text; it’s not specific.*)

The shared-task dataset includes 186 argumentative passages and 4,136 total questions. Of those questions, 67% are labeled **USEFUL**, 22% **UNHELPFUL**, and 11% **INVALID**. For our experiments, we allocate 80% of the data for training (when needed) and reserve the remaining 20% (38 passages) for testing to ensure comparability across models.

3.2 Evaluation

We follow the official shared-task protocol: for each intervention, a system generates three critical questions. Each generated question is matched to the nearest reference using semantic similarity, with a threshold of 0.6, and then assigned one of three labels (**USEFUL**, **UNHELPFUL**, **INVALID**). We award 0.33 points for each **USEFUL** question and zero points for **UNHELPFUL** or **INVALID** questions, yielding a final score in [0,1] (e.g., three **USEFUL** → 1.0; two → 0.66; one → 0.33; none → 0.0).

Because the reference set is not exhaustive, many generated questions fall below the similarity thresh-

Table 1: Accuracy, Precision, Recall and F1-score for each model, for classifying the *useful* label.

Model	Accuracy	Precision	Recall	F1-score
Llama-3.3-70B	71.35	71.47	95.73	81.84
DeepSeek-V3	72.24	72.93	93.45	81.80
Llama-3.1-8B	68.27	68.82	97.77	80.78
Qwen2.5-72B	66.61	75.95	73.91	74.91

old and remain unlabeled. To handle these cases, we perform a secondary LLM-based classification over all 4,136 questions. Table 1 reports the Accuracy, Precision, Recall, and F1-score for identifying the USEFUL label; Llama-3.3-70B achieves the highest F1-score of 81.84%. Going forward, we present results both with and without this LLM-driven fallback for below-threshold questions.

For full LLM evaluation details, see Appendix B.

3.3 Benchmarking

We evaluated ten models (ranging from 3B to 405B parameters), including one reasoning-tuned model. Across three families:

- **Llama-3** (Dubey et al., 2024) (3.1-405B, 3.3-70B, 3.1-8B, 3.2-3B)
- **DeepSeek** (R1, V3-0324 (DeepSeek-AI et al., 2025))
- **Qwen-2.5** (Qwen et al., 2025) (72B, 14B, 7B, 3B)

Each model was tested with five prompting strategies:

- **ZeroShot**: “Suggest 3 critical questions.”
- **MinimalZeroShot**: ZeroShot plus a brief note on what makes a question critical.
- **BasicCoT**: MinimalZeroShot with a “think step-by-step” instruction.
- **FewShot**: Incorporates a few annotated examples.
- **FewShotFullCoT**: Full guidelines with examples and explanations for UNHELPFUL and INVALID classifications.

Table 2 reports, for each model, the mean $\pm$ SD of (1) punctuation score, (2) punctuation score with LLM labels (for questions below the similarity threshold), and (3) percentage of LLM-labeled questions. Rows are sorted by Mean Punc. w/ LLM

(highest to lowest). Llama-3 (3.1-405B) leads with a Mean Punc. w/ LLM of 0.851, and overall the FewShot and FewShotFullCoT prompts perform best. Full prompting details and per-prompt results are in Appendix C.

3.4 Reinforcement Learning

We adapt *Group-Relative Policy Optimisation* (GRPO) (Shao et al., 2024) to CQG. Every iteration samples *eight* candidate generations per prompt, uses group-normalised rewards for outcome supervision, and applies a batch-level KL penalty to keep the new policy close to the old one.

Group-relative outcome supervision. For an intervention q , we draw a *group* of G candidate outputs from the *old* policy $\pi_{\theta_{\text{old}}}$,

$$\mathcal{O}(q) = \{o_1, \dots, o_G\}, \quad o_i \sim \pi_{\theta_{\text{old}}}(\cdot | q).$$

A reward model scores each output, $r = \{r_1, \dots, r_G\}$, after which the scores are *group-normalised*

$$\hat{r}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r)}.$$

This normalised reward is treated as a constant token-level advantage: $\hat{A}_{i,t} = \hat{r}_i$ for every token position t in o_i . The policy is then updated to increase the expected advantage while softly constraining the KL divergence from the old policy.

Composite reward. Our final reward is $R = R_{\text{fmt}} + R_{\text{struct}} + R_{\text{corr}}$:

- **Formatting** R_{fmt} checks the required XML tags `<reasoning> . . . </reasoning>` `<answer> . . . </answer>` and grants $\{0, 0.5, 1.0, 1.5\}$ points.
- **Structure** R_{struct} gives 0.5 if *exactly* three CQs appear inside `<answer>`, else 0.
- **Correctness** R_{corr} awards $\{0, 0.5, 1.0, 1.5\}$ for $\{0, 1, 2, 3\}$ valid CQs, using sentence-BERT similarity with a Llama-3.3-70B judge as fallback when similarity is below threshold.

Fine-tuning details. Rank-64 LoRA adapters are inserted into seven projection layers of every transformer block (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj). GRPO is run for 3–7 epochs on two base models: Qwen-2.5-3B (Qwen et al., 2025) and Llama-3.1-8B (Dubey et al., 2024), updating only the LoRA weights.

Table 2: Mean $\pm$ SD of (1) punctuation score, (2) punctuation score with LLM labels (for questions below the similarity threshold), and (3) LLM-labeled percentage, along with each model’s best prompt, , sorted by Mean Punc. w/ LLM from highest to lowest.

Model	Mean Punc.	Mean Punc. w/ LLM	Mean LLM-labeled %	Best Prompt
Llama-3.1-405B	0.763 $\pm$ 0.045	0.851 $\pm$ 0.041	8.94% $\pm$ 2.15%	FewShot
DeepSeek R1	0.597 $\pm$ 0.050	0.840 $\pm$ 0.013	24.38% $\pm$ 4.75%	FewShotFullCoT
Llama-3.3-70B	0.730 $\pm$ 0.058	0.839 $\pm$ 0.078	11.04% $\pm$ 2.60%	FewShotFullCoT
Llama-3.1-8B	0.702 $\pm$ 0.045	0.826 $\pm$ 0.031	12.62% $\pm$ 3.04%	FewShot
DeepSeek-V3-0324	0.686 $\pm$ 0.049	0.824 $\pm$ 0.022	14.40% $\pm$ 4.55%	FewShot
Qwen2.5-72B	0.731 $\pm$ 0.029	0.800 $\pm$ 0.034	7.18% $\pm$ 2.88%	BasicCoT
Llama-3.2-3B	0.590 $\pm$ 0.131	0.779 $\pm$ 0.134	19.28% $\pm$ 6.39%	FewShot
Qwen2.5-14B	0.602 $\pm$ 0.141	0.742 $\pm$ 0.164	15.08% $\pm$ 4.60%	FewShot
Qwen2.5-7B	0.600 $\pm$ 0.142	0.767 $\pm$ 0.162	18.08% $\pm$ 6.17%	FewShot
Qwen2.5-3B	0.486 $\pm$ 0.122	0.698 $\pm$ 0.156	26.50% $\pm$ 8.50%	FewShot

4 Results and Discussion

Table 2 reports punctuation scores before and after RL fine-tuning. GRPO yields consistent gains on Qwen2.5-3B (+0.128 absolute with LLM labels) and improves the Llama-3.1-8B metric at three epochs (+0.025), but longer training over-fits the sparse reward and degrades performance. Interestingly, RL reduces the share of LLM-labelled outputs, suggesting better coverage of the reference set. Overall, policy optimisation helps, yet the noisy reward signal—ultimately tied to subjective usefulness—limits further progress.

Training dynamics (Figure 1). Both models learn the reward signal quickly: within the first 300 steps the *total* reward rises from $\sim$ 2.0 to just under 3.0, close to the 3.5 upper bound. Most of the gain comes from better formatting and structure (Fig. 1a vs. c), showing that GRPO enforces hard constraints well.

The *correctness* curves (panels b,d) reveal an interesting asymmetry. Llama-3.1-8B *starts* at a higher correctness score ($\sim$ 1.1) than Qwen-2.5-3B ($\sim$ 0.7), so its headroom for improvement is smaller. After three epochs it hovers around 1.28—roughly two valid CQs per sample—whereas Qwen climbs more steeply from a lower baseline and converges to a similar level. One likely explanation is model capacity: the larger Llama already possesses stronger reasoning ability, so LoRA rank 64 may not provide enough adaptation capacity, or more training steps may be required to move off a good-but-safe local optimum.

After 600–700 steps (5 epochs) both models flatten and variance increases, which matches Table 2: longer training no longer helps and may even hurt.

Overall, our composite reward is sufficient for basic compliance, but a richer (preference-based) signal—and perhaps higher-rank adapters for larger backbones—is needed to push correctness beyond the two-out-of-three “good-enough” regime.

Discussion (Table 3). RL fine-tuning reshuffles the leaderboard in two ways. First, Qwen-2.5-3B leaps from the bottom to mid-pack, confirming that small models benefit the most from the composite reward. Second, Llama-3.1-8B ties the 405 B titan after only three epochs—yet it requires more LLM labelling (18 % vs 9 %), showing that the larger backbone still generates a non-trivial share of below-threshold CQs. Longer RL (5–7 epochs, Table 2) degrades both models, mirroring the plateau in Figure 1. Overall, RL narrows the gap between small and very-large models, but fully closing it may need richer preference feedback or higher-rank adapters for bigger backbones. These jumps validate the *potential* of GRPO+LoRA for CQG: even with rank-64 adapters and modest training budgets, we nearly match the 405 B frontier. We therefore conjecture that scaling the same recipe to larger backbones (e.g. 70 B+), higher-rank adapters, and longer RL schedules could push punctuation scores toward the theoretical maximum, provided a richer preference-based reward is available.

The rest of the reward functions are discussed in Appendix D.

5 Limitations

Subjectivity of “criticality”. Whether a question genuinely challenges an argument often depends on the reader’s prior knowledge and values. Two annotators may disagree on the same CQ—even

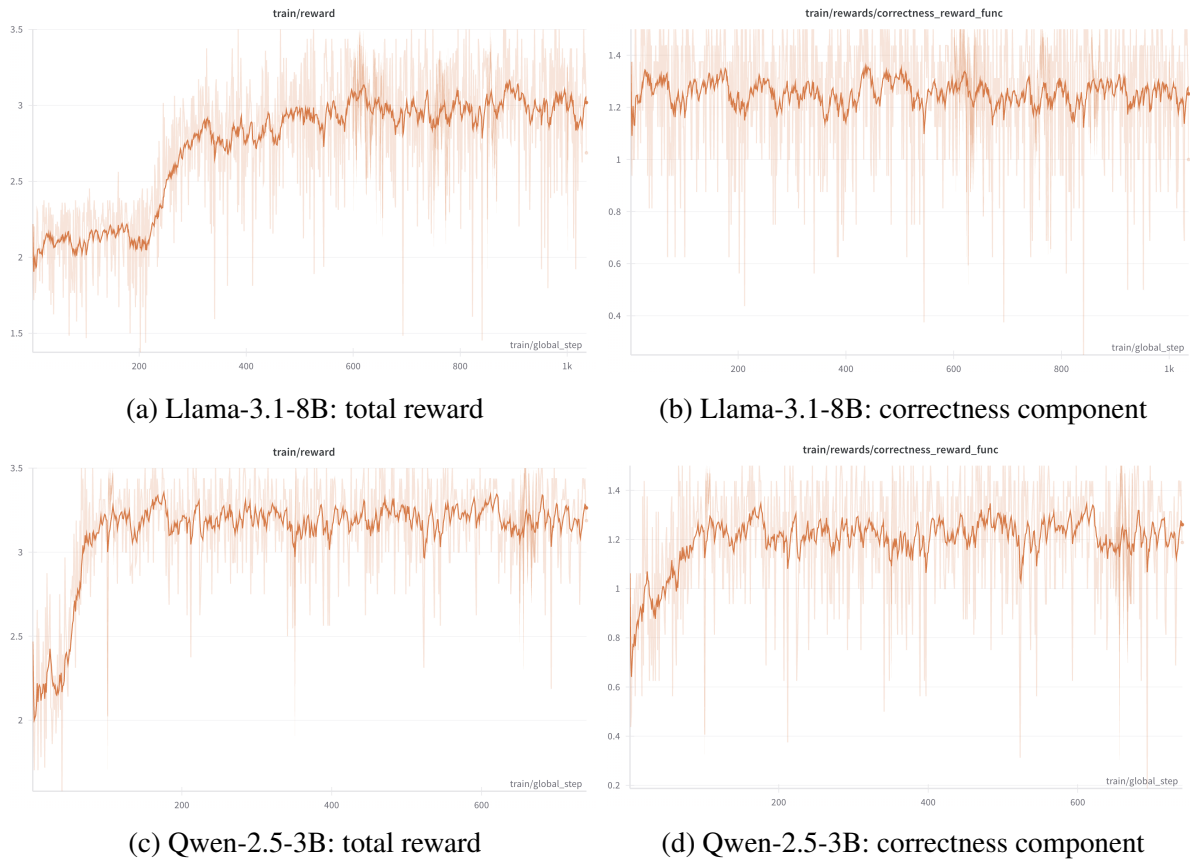


Figure 1: Reward curves during GRPO fine-tuning. Shaded areas show per-batch variability; dark lines are EMA-smoothed means. The theoretical maximum is 3.5 ($=1.5$ format + 0.5 structure + 1.5 correctness).

after reading identical guidelines—leading to noisy gold labels.

Evaluation gap. The shared-task metric combines semantic similarity with a three-way label lookup; anything falling below a cosine threshold is simply ignored. Our own fallback (LLM grading) raises coverage, but its agreement with humans remains imperfect and expensive to scale. A more reliable protocol—e.g. human pairwise preference or rubric-based scoring—remains an open problem.

Prompt sensitivity. All three of our systems rely on carefully engineered prompts. Minor phrasing changes can shift the question type distribution, making reproducibility hard and robustness fragile.

Training data scarcity. Only 186 interventions are available for supervision. Coverage of rare schemes and domain-specific topics is limited, which likely undercuts RL fine-tuning.

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Table 3: Updated leaderboard after RL fine-tuning. We report mean $\pm$ SD punctuation, punctuation with LLM labels (for questions below the similarity threshold), and the percentage of LLM-labelled questions. Rows are sorted by **Punc. w/ LLM**.

Model	Punc.	Punc. w/ LLM	LLM-labeled %
Llama-3.1-405B	0.763 ± 0.045	0.851 ± 0.041	8.94% $\pm 2.15\%$
Llama-3.1-8B-w/RL (3 epochs)	0.667	0.851	18.42%
DeepSeek R1	0.597 ± 0.050	0.840 ± 0.013	24.38% $\pm 4.75\%$
Llama-3.3-70B	0.730 ± 0.058	0.839 ± 0.078	11.04% $\pm 2.60\%$
Llama-3.1-8B	0.702 ± 0.045	0.826 ± 0.031	12.62% $\pm 3.04\%$
DeepSeek-V3-0324	0.686 ± 0.049	0.824 ± 0.022	14.40% $\pm 4.55\%$
Qwen2.5-3B-w/RL (3 epochs)	0.614	0.816	20.17%
Qwen2.5-72B	0.731 ± 0.029	0.800 ± 0.034	7.18% $\pm 2.88\%$
Llama-3.2-3B	0.590 ± 0.131	0.779 ± 0.134	19.28% $\pm 6.39\%$
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Qwen2.5-7B	0.600 ± 0.142	0.767 ± 0.162	18.08% $\pm 6.17\%$
Qwen2.5-3B	0.486 ± 0.122	0.698 ± 0.156	26.50% $\pm 8.50\%$

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A Human Evaluation Guidelines

This is the full annotations guidelines as provided by the shared task ([hitz-zentroa, 2025](#)):

Critical Questions Evaluation Guidelines:

In this assignment, you should imagine that you are a teacher in a class on critical thinking. During this class, your goal is to get students to critically analyze the argumentative texts presented to them and provide them with the tools to evaluate the validity of the arguments in the text. To achieve this goal, students will be given a paragraph with argumentative content and you should pose questions to them about this paragraph. You should read the paragraph, look at the questions we suggest, and then assess which of these questions might be useful to the students and which ones might not.

For each of the questions you will have to decide if it is *useful*, *unhelpful*, or if it is not even *valid* for the arguments of that paragraph. Here we define these categories: :

Classification Categories

1. **Useful question** One should not take the arguments in the text as valid without having reflected on this question.
2. **Unhelpful question** The question makes sense, is related to the text and in some exceptional cases the answer could invalidate or diminish the validity of the arguments in the text but, as a teacher, you would probably not suggest it. This may be in cases where:
 - (a) The answer to the question is common sense.
 - (b) The answer to the question is a well-known fact that does not generate controversy.
 - (c) The question is very complicated to understand and it would be impractical to question the arguments.
 - (d) The question is answered in the text itself.
3. **Invalid question** A question is invalid when the answer to this question cannot serve to invalidate or diminish the acceptability of the arguments of the text. This can be for several reasons:

- (a) Unrelated: The question is unrelated to the text.
- (b) New concept: The question introduces new concepts that were not in the text.
- (c) Bad reasoning: The question does not challenge any argument defended in the text. For example, when the question challenges the opposite position to the one defended in the text.
- (d) Very general: The question is very vague and does not ask about anything specific in the text. This question could be asked of any argument.
- (e) Non-critical: Although the question asks about something in the text, it is not critical of any argument. For example, when the question is a reading-comprehension one. **A question is only critical if the answer to the question can potentially reduce the validity of the argument.**

As follows, you can see a couple of texts in which the questions have already been classified:

Examples

Example 1

Text:

MT: "We have indeed required those products to have very large warnings on them, telling people about their salt and fat content. And we don't allow them to say things which are misleading. We don't allow fatty products to say they're healthy. I've got in front of me a letter from a credit card company. There are two figures on this letter which really stand out: 0% per annum new balance transfers, 0% per annum new money transfers. Now, underneath that, in small print, it becomes absolutely clear that you'll be paying all sorts of charges. And then very, very quickly you'll be lured into paying a very large charge. Do you think it's responsible to encourage people to mis-read something like that? I've got all the information there in front of me."

Useful Questions

- What is the basis for comparing the credit card company's advertisement to food products, and is this comparison valid?

- Is it actually the case that fatty products that have potential bad consequences are not allowed to be described in misleading ways? Is there evidence for this claim?
- Are there other factors in this particular case that could have interfered with the event of 'people being lured into paying a very large charge'?
- Does the credit card company's offer of 0% interest rates for new balances and transfers accurately reflect the true cost of borrowing? What hidden fees or charges might consumers face?
- How does the mentioned credit card offer differ between its initial presentation and fine print details?
- What specific charges does the credit card company advertise as being 0% per annum, and how do these charges compare to the charges that will actually be paid?

Unhelpful Questions

- How does the speaker know that the credit card company's advertisement is irresponsible, and what evidence does the speaker have to support this claim? → *it says it in the text*
- How might the credit card company's advertisement be improved to make it more responsible and less misleading, and what changes would need to be made to achieve this? → *answering this is very complicated and not to the point*
- How strong is the generalization that if all sorts of charges are written in small print then people will be lured into paying a very large charge? → *to some extent, this is common sense*
- What are the potential consequences of the credit card company's advertisement being misleading, and how might these consequences affect consumers? → *the answer to this question is most likely going to support the argument, not diminish its validity*
- What assumptions does MT make when arguing against allowing fatty products to claim they are healthy? Are these assumptions justified? → *this question is very broad and difficult to answer*
- What evidence does the speaker have to support their claim that the credit card company's practices are irresponsible? → *the text already gives these reasons*

Invalid Questions

- Are there any assumptions made by the speaker that are not explicitly stated but are implied by their argument? → *this question could be raised for any text, it's not specific*
- Are these measures effective in preventing misleading claims by manufacturers? → *no measures are being proposed in this text*
- Is the speaker's argument based on a logical chain of reasoning, or are there any gaping holes or flaws in the argument? → *this question could be raised for any text, it's not specific*
- What measures has MT taken to ensure accurate labeling and advertising of food products with regards to salt and fat content? → *MT did not take any measures themself*

Example 2

Text:

JW: "Well, debt is morally neutral in the sense that it can be good to be in debt. Sometimes it's a good idea, and sometimes it's a bad idea. And so I think there's been broad agreement about that in the discussion so far. The question is why is there too much debt? Why are people taking on too much debt? Now, a lot of people come up with rather glib moral answers: somehow bankers want them to get into excessive debt, which is a very peculiar idea, when you think that it's the bankers who aren't going to get repaid. So it's a very odd idea. There's too much debt for the same reason that there are too many unmarried mothers, and for the same reason that you get an overproduction of sheep."

Useful Questions

- If debt is considered 'morally neutral', then how do we determine whether someone's level of debt is acceptable or not? Are there specific circumstances under which debt becomes immoral or irresponsible?

- What is being defined as ‘too much debt’?
- How does the concept of ‘excessive debt’ differ from simply having debt? In other words, at what point does debt become problematic?
- How does the comparison to unmarried mothers and overproduction of sheep relate to the issue of debt?

Unhelpful Questions

- What evidence supports that it is generally accepted that debt is morally neutral in the sense that it can be good to be in debt? → *this question is too complicated to answer*
- What are the potential consequences of labeling debt as “good” or “bad”? → *this question is too complicated to answer*
- What are the potential implications of the author’s perspective on the issue of debt for society as a whole? → *this question is too complicated to answer*

Invalid Questions

- In what ways is debt considered morally neutral according to JW’s perspective? Can you provide examples to illustrate this point? → *invalid for 2 reasons: 1. it’s asking about the speakers’ perspective, 2. it’s asking the user (the students) to provide examples*
- How does JW suggest we approach understanding why there is too much debt in society? Does he offer any alternative perspectives beyond those already discussed? → *this is a reading-comprehension question*
- Why might banks encourage people to take on more debt than they can afford? Are they acting in their own self-interest, or are there broader societal forces at play? → *the text does not say that banks encourage people to take more debt*
- Could you summarize JW’s overall stance on debt and how it differs from more traditional views presented earlier? → *reading-comprehension question, not critical towards any argument*
- What factors are leading people to take on more debt than they should? Are they making poor financial decisions, facing economic

pressures, responding to marketing tactics, or something else entirely? → *the first part of this question is valid, but the second one, suggesting answers, is providing new concepts: “making poor financial decisions, facing economic pressures, responding to marketing tactic” not present in the original text*

- How does the issue of ‘too much debt’ relate to larger social and economic trends, such as income inequality, globalization, or technological change? → *the question is introducing new concepts not in the text: “income inequality, globalization, or technological change”*

Example 3

Text:

JL: "So, if I want a washing machine, I have to go to the launderette every week at the moment and spend several pounds in the launderette. But, if I buy the washing machine, on credit, I can have my own washing machine, for pretty much the same amount of money that I’m spending in the launderette. That seems to me to be beneficial. It raises living standards. And this idea of usury has been—well, it’s been fiddled, hasn’t it, over the years? Because why did we have money lenders from one religious persuasion unable to lend to their own persuasion, but happily would lend to those of another religious persuasion? I don’t buy that at all. I think that credit raises living standards. It’s beneficial to people because it allows them to fund today’s purchases out of their future income."

Helpful Questions

- What are the long-term implications of taking on debt to purchase a washing machine? How might this affect the individual’s financial situation in the future?
- How does the argument that buying a washing machine on credit allows people to fund today’s purchases out of their

future income address the potential drawbacks or risks of taking on debt?

- What is the basis for the claim that credit raises living standards? Is there evidence to support this claim?
- If buying a washing machine on credit, will living standards rise? What evidence supports this claim? How likely are the consequences?
- Is the comparison between the cost of using a launderette and the cost of buying a washing machine on credit accurate? Are there other factors to consider?
- What other consequences should also be taken into account if people can fund today's purchases out of their future income?
- Are there any other factors that should be considered when deciding whether or not to buy a washing machine on credit?

Unhelpful Questions

- What are the assumptions underlying the idea that buying a washing machine on credit is a good idea? → *this question is complicated because it's too broad*

Not Valid Questions

- What are some potential drawbacks or negative consequences of relying heavily on credit to raise living standards, as suggested by JL's argument? → *the question is a reading-comprehension one, as indicated by the cue "as suggested by JL"*
- Values: Improving living standards through access to consumer goods. Justified? Debatable, as some argue that excessive debt and consumption can lead to financial instability and other negative consequences → *this is not a question*
- Is the reasoning valid? (a) Are there any logical errors or fallacies present? (b) Can you explain how the speaker arrived at the conclusion based on the given facts? → *this question could be raised for any text, it's not specific*
- Are the premises true and relevant to the conclusion? (a) Is there enough evidence provided to support each statement made? (b) Do the statements actually

lead to the conclusion drawn? → *this question could be raised for any text, it's not specific*

- What is the speaker's understanding of the history of money lending and the religious restrictions that have been placed on it? How does the speaker account for the fact that some religious traditions have prohibited usury while others have allowed it? → *this question is not critical with the text, it's merely asking about the content*
- How does the speaker justify the idea that credit is beneficial to people? What specific benefits does the speaker believe credit provides? → *this question is not critical with the text, it's merely asking about the content*

B LLM evaluation details

We benchmarked four models: DeepSeek V3 (DeepSeek-AI et al., 2025), Llama-3.3-70B and Llama-3.1-8B (Dubey et al., 2024), and Qwen 2.5-72B (Qwen et al., 2025). All models were loaded from HuggingFace (Wolf et al., 2020) and queried either with vLLM or through DeepInfra's API.¹

Every model was queried with a temperature of 0, and the prompt is the full guidelines provided in Appendix A, with the addition of the following:

¹<https://deepinfra.com/>

Your task is to classify each suggested question as useful, unhelpful, or invalid, and provide a brief justification for your classification.

Evaluation Task: Now, based on the guidelines above, you will be given an intervention (a paragraph from an argumentative text) and a question that attempts to challenge it. Your task is to classify the question into one of the three categories: USEFUL, UNHELPFUL, or INVALID. Provide the label in the following format:

Intervention:
{intervention}

Question:
{cq}
ANSWER:

C Benchmarking details

All models were loaded from HuggingFace (Wolf et al., 2020) and queried either with vLLM or through DeepInfra’s API.² Every model was queried with a temperature of 0, and the prompts are as follows:

ZeroShot prompt

Suggest 3 critical questions that should be raised before accepting the arguments in this text:
{intervention}
Give one question per line. Make the questions simple, and do not provide any explanation regarding why the question is relevant.
Do not include any special characters or numbering except for the question mark.

MinimalZeroShot

You are a teacher in a critical thinking class. Your goal is to help students learn to critically evaluate argumentative texts.

To do this, you need to generate critical questions that challenge the validity of the arguments presented. A question is considered USEFUL if it makes the reader reflect on the text in a way that could potentially diminish its perceived validity. Avoid questions that are common sense, reading-comprehension, too general, or that introduce new concepts not present in the text.

Guidelines:

1. USEFUL QUESTION:

- Should be raised before accepting the arguments in this text.
- Challenges the text’s argument in a meaningful way.
- Prompts critical reflection that can weaken the argument’s validity if answered.
- Focuses on details already present in the text without introducing external ideas.

2. UNHELPFUL QUESTION:

- Although related to the text, it asks about aspects that are either common sense, well-known facts, or too complicated.

3. INVALID QUESTION:

- Unrelated to the text or introduces new concepts.
- Uses vague language or fails to challenge the argument’s core reasoning.

Text:{intervention}

Provide one question per line. Do not include any special characters or numbering except for the question mark.

²<https://deepinfra.com/>

BasicCoT

You are a teacher in a critical thinking class. Your goal is to help students learn to critically evaluate argumentative texts.

To do this, you need to generate critical questions that challenge the validity of the arguments presented. A question is considered USEFUL if it makes the reader reflect on the text in a way that could potentially diminish its perceived validity. Avoid questions that are common sense, reading-comprehension, too general, or that introduce new concepts not present in the text.

Guidelines:

1. USEFUL QUESTION:

- Should be raised before accepting the arguments in this text.
- Challenges the text's argument in a meaningful way.
- Prompts critical reflection that can weaken the argument's validity if answered.
- Focuses on details already present in the text without introducing external ideas.

2. UNHELPFUL QUESTION:

- Although related to the text, it asks about aspects that are either common sense, well-known facts, or too complicated.

3. INVALID QUESTION:

- Unrelated to the text or introduces new concepts.
 - Uses vague language or fails to challenge the argument's core reasoning.
- Now, using the guidelines above, generate three USEFUL critical questions for the following text:

TEXT:{intervention}

Let's think step by step, Respond in the following format:

<reasoning>

nYour reasoning here.

n</reasoning>

n<answer>

nProvide three questions one question per line. Do not include any special characters or numbering except for the question mark.

n</answer>"

FewShot

You are a teacher in a critical thinking class. Your goal is to help students learn to critically evaluate argumentative texts. To do this, you need to generate critical questions that challenge the validity of the arguments presented. A question is considered USEFUL if it makes the reader reflect on the text in a way that could potentially diminish its perceived validity. Avoid questions that are common sense, reading-comprehension, too general, or that introduce new concepts not present in the text.

Guidelines: 1. USEFUL QUESTION:

- Challenges the text's argument in a meaningful way.
- Prompts critical reflection that can weaken the argument's validity if answered.
- Focuses on details already present in the text without introducing external ideas.

2. UNHELPFUL QUESTION:

- Although related to the text, it asks about aspects that are either common sense, well-known facts, or too complicated.

3. INVALID QUESTION:

- Unrelated to the text or introduces new concepts.
- Uses vague language or fails to challenge the argument's core reasoning.

Few-Shot Examples:

Example 1

TEXT: MT: We have indeed required those products to have very large warnings on them, telling people about their salt and fat content And we don't allow them to say things which are misleading We don't allow fatty products to say they're healthy I've got in front of me a letter from a credit card company There are two figures on this letter which really stand out: 0% per annum new balance transfers, 0% per annum new money transfers Now, underneath that, in small print, it becomes absolutely clear that you'll be paying all sorts of charges And then very, very quickly you'll be lured into paying a very large charge Do you think it's responsible to encourage people to misread something like that?

I've got all the information there in front of me Why is that irresponsible?

Useful Questions:

- What is the basis for comparing the credit card company's advertisement to food products, and is this comparison valid?
- Is there evidence that prevents misleading descriptions of fatty products, and does the text support this claim?
- Does the credit card company's offer of 0% interest truly reflect the cost of borrowing, given the hidden fees?

Unhelpful Question:

- How does the speaker know that the advertisement is irresponsible? (This is already stated in the text.)

Invalid Question: - Are there any assumptions made by the speaker that are not explicitly stated? (Too generic.)

Now, using the guidelines and examples above, generate three USEFUL critical questions for the following text:

TEXT: {intervention}

Provide one question per line. Do not include any special characters or numbering except for the question mark.

FewShotFullCoT

You are a teacher in a critical thinking class. Your goal is to help students learn to critically evaluate argumentative texts. To do this, you need to generate critical questions that challenge the validity of the arguments presented. A question is considered USEFUL if it makes the reader reflect on the text in a way that could potentially diminish its perceived validity. Avoid questions that are common sense, reading-comprehension, too general, or that introduce new concepts not present in the text.

Guidelines:

1. USEFUL QUESTION:

- Challenges the text's argument in a meaningful way.
- Prompts critical reflection that can weaken the argument's validity if answered.
- Focuses on details already present in the text without introducing external ideas.

2. UNHELPFUL QUESTION:

- Although related to the text, it asks about aspects that are either common sense, well-known facts, or too complicated.
- May lead to answers that do not diminish the argument's validity.

3. INVALID QUESTION:

- Unrelated to the text or introduces new concepts not present in the text.
- Uses vague language or fails to challenge the argument's core reasoning.

Few-Shot Examples:

Example 1

TEXT

MT: We have indeed required those products to have very large warnings on them, telling people about their salt and fat content. And we don't allow them to say things which are misleading. We don't allow fatty products to say they're healthy. I've got in front of me a letter from a credit card company. There are two figures on this letter which really stand out: 0% per annum new balance transfers, 0% per annum new money transfers. Now, underneath that, in small print, it becomes absolutely clear that you'll be paying all sorts of charges

And then very, very quickly you'll be lured into paying a very large charge. Do you think it's responsible to encourage people to mis-read something like that? I've got all the information there in front of me.

USEFUL QUESTIONS

- What is the basis for comparing the credit card company's advertisement to food products, and is this comparison valid?
- Is it actually the case that fatty products that have potential bad consequences are not allowed to be described in misleading ways? Is there evidence for this claim?
- Are there other factors in this particular case that could have interfered with the event of 'people being lured into paying a very large charge'?
- Does the credit card company's offer of 0% interest rates for new balances and transfers accurately reflect the true cost of borrowing? What hidden fees or charges might consumers face?
- How does the mentioned credit card offer differ between its initial presentation and fine print details?
- What specific charges does the credit card company advertise as being 0% per annum, and how do these charges compare to the charges that will actually be paid?

UNHELPFUL QUESTIONS

- How does the speaker know that the credit card company's advertisement is irresponsible, and what evidence does the speaker have to support this claim? → it says it in the text
- How might the credit card company's advertisement be improved to make it more responsible and less misleading, and what changes would need to be made to achieve this? → answering this is very complicated and not to the point
- How strong is the generalization that if all sorts of charges are written in small print then people will be lured into paying a very large charge? → to some extent, this is common sense

- What are the potential consequences of the credit card company's advertisement being misleading, and how might these consequences affect consumers? → the answer to this question is most likely going to support the argument, not diminish its validity
- What assumptions does MT make when arguing against allowing fatty products to claim they are healthy? Are these assumptions justified? → this question is very broad and difficult to answer
- What evidence does the speaker have to support their claim that the credit card company's practices are irresponsible? → the text already gives these reasons

INVALID QUESTIONS

- Are there any assumptions made by the speaker that are not explicitly stated but are implied by their argument? → this question could be raised for any text, it's not specific
- Are these measures effective in preventing misleading claims by manufacturers? → no measures are being proposed in this text
- Is the speaker's argument based on a logical chain of reasoning, or are there any gaping holes or flaws in the argument? → this question could be raised for any text, it's not specific
- What measures has MT taken to ensure accurate labeling and advertising of food products with regards to salt and fat content? → MT did not take any measures themselves

Example 2

TEXT JW: Well, debt is morally neutral in the sense that it can be good to be in debt. Sometimes it's a good idea, and sometimes it's a bad idea. And so I think there's been broad agreement about that in the discussion so far. The question is why is there too much debt? Why are people taking on too much debt? This one is the elaboration of the previous question. Now, a lot of people come up with rather glib moral answers: somehow bankers want them to get into excessive debt which is a very peculiar idea, when you think that it's the bankers who aren't going to get repaid.

So it's a very odd idea there's too much debt for the same reason that there are too many unmarried mothers, and for the same reason that you get an overproduction of sheep.

USEFUL QUESTIONS

- If debt is considered 'morally neutral', then how do we determine whether someone's level of debt is acceptable or not? Are there specific circumstances under which debt becomes immoral or irresponsible?
- What is being defined as 'too much debt'?
- How does the concept of 'excessive debt' differ from simply having debt? In other words, at what point does debt become problematic?

- How does the comparison to unmarried mothers and overproduction of sheep relate to the issue of debt?

UNHELPFUL QUESTIONS

- What evidence supports that it is generally accepted that debt is morally neutral in the sense that it can be good to be in debt? → this question is too complicated to answer
- What are the potential consequences of labeling debt as good or bad? → this question is too complicated to answer
- What are the potential implications of the author's perspective on the issue of debt for society as a whole? → this question is too complicated to answer

INVALID QUESTIONS

- In what ways is debt considered morally neutral according to JW's perspective? Can you provide examples to illustrate this point? → invalid for 2 reasons: 1. it's asking about the speakers' perspective, 2. it's asking the user (the students) to provide examples
- How does JW suggest we approach understanding why there is too much debt in society? Does he offer any alternative perspectives beyond those already discussed in the conversation? → this is a reading-comprehension question

- Why might banks encourage people to take on more debt than they can afford? Are they acting in their own self-interest, or are there broader societal forces at play here? → the text does not say that banks encourage people to take more debt

- Could you summarize JW's overall stance on debt and how it differs from more traditional views presented earlier in the discussion? → reading-comprehension question, not critical towards any argument

- What factors are leading people to take on more debt than they should? Are they making poor financial decisions, facing economic pressures, responding to marketing tactics, or something else entirely? → the first part of this question is valid, but the second one, suggesting answers, is providing new concepts not present in the original text

- How does the issue of 'too much debt' relate to larger social and economic trends, such as income inequality, globalization, or technological change? → the question is introducing new concepts not in the text: income inequality, globalization, or technological change

Example 3

TEXT

JL: Šo, if I want a washing machine, I have to go to the launderette every week at the moment and spend several pounds in the launderette. But, if I buy the washing machine, on credit, I can have my own washing machine, for pretty much the same amount of money that I'm spending in the launderette. That seems to me to be beneficial. It raises living standards.

And this idea of usury has been - well, it's been fiddled, hasn't it, over the years? Because why did we have money lenders from one religious persuasion unable to lend to their own persuasion, but happily would lend to those of another religious persuasion? I don't buy that at all. I think that credit raises living standards. It's beneficial to people because it allows them to fund today's purchases out of their future income.

HELPFUL QUESTIONS

- What are the long-term implications of taking on debt to purchase a washing machine? How might this affect the individual's financial situation in the future?
- How does the argument that buying a washing machine on credit allows people to fund today's purchases out of their future income address the potential drawbacks or risks of taking on debt?
- What is the basis for the claim that credit raises living standards? Is there evidence to support this claim?
- If buying a washing machine on credit, will living standards rise? What evidence supports this claim? How likely are the consequences?
- Is the comparison between the cost of using a launderette and the cost of buying a washing machine on credit accurate? Are there other factors to consider?
- What other consequences should also be taken into account if people can fund today's purchases out of their future income?
- Are there any other factors that should be considered when deciding whether or not to buy a washing machine on credit?

UNHELPFUL QUESTIONS

- What are the assumptions underlying the idea that buying a washing machine on credit is a good idea? → this question is complicated because it's too broad NOT VALID
- What are some potential drawbacks or negative consequences of relying heavily on credit to raise living standards, as suggested by JL's argument? → the question is a reading-comprehension one, as indicated by the cue "as suggested by JL"
- [Additional non-question statements are not valid: e.g., 'Values: Improving living standards through access to consumer goods. Justified? Debatable, as some argue that excessive debt and consumption can lead to financial instability and other negative consequences' is not a question]

- Is the reasoning valid? (a) Are there any logical errors or fallacies present? (b) Can you explain how the speaker arrived at the conclusion based on the given facts? → this question could be raised for any text, it's not specific
 - Are the premises true and relevant to the conclusion? (a) Is there enough evidence provided to support each statement made? (b) Do the statements actually lead to the conclusion drawn? → this question could be raised for any text, it's not specific
 - What is the speaker's understanding of the history of money lending and the religious restrictions that have been placed on it? How does the speaker account for the fact that some religious traditions have prohibited usury while others have allowed it? → this question is not critical with the text, it's merely asking about the content
 - How does the speaker justify the idea that credit is beneficial to people? What specific benefits does the speaker believe credit provides? → this question is not critical with the text, it's merely asking about the content
- Now, using the guidelines and examples above, generate three **USEFUL** critical questions for the following text:

TEXT:

{intervention}

Provide one question per line. Do not include any special characters or numbering except for the question mark."

The results for each model per prompt is shown in Table 4

D Reward-Component Curves

Observations.

- **Llama-3.1-8B**—Panels (a–d). Soft formatting is mastered sharply at step ~230, but strict regex compliance improves more slowly and never fully reaches the 0.5 ceiling, indicating residual minor inflections. The **STRUCTURE** reward saturates almost immediately, while **XML-**

Table 4: Punctuation results for each model across prompting methods. Each cell reports the triplet: (1) punctuation score, (2) punctuation when using LLM labels (for questions below the similarity threshold), and (3) the LLM-labeled percentage. **Boldface** marks the best prompt for each model (top value in each row), and underlining marks the top score in each column (best model per prompt).

Model	ZeroShot	MinimalZeroShot	BasicCoT	FewShot	FewShotFullCoT
DeepSeek R1	0.535 – 0.842 – 30.7%	0.553 – 0.825 – 27.2%	<u>0.623 – 0.860 – 23.7%</u>	0.623 – 0.842 – 21.9%	0.649 – 0.833 – 18.4%
DeepSeek-V3-0324	0.711 – 0.816 – 12.3%	0.640 – 0.842 – 21.1%	0.632 – 0.789 – 15.8%	0.702 – 0.842 – 14.0%	0.746 – 0.833 – 8.8%
Qwen2.5-72B	0.684 – 0.754 – 7.9%	0.754 – 0.825 – 7.0%	0.728 – 0.842 – 11.4%	0.754 – 0.789 – 3.5%	0.737 – 0.789 – 6.1%
Qwen2.5-14B	0.588 – 0.719 – 17.5%	0.368 – 0.465 – 10.5%	0.632 – 0.842 – 21.1%	0.693 – 0.851 – 15.8%	0.728 – 0.833 – 10.5%
Qwen2.5-7B	0.518 – 0.772 – 28.1%	0.395 – 0.491 – 11.4%	0.649 – 0.816 – 18.4%	<u>0.746 – 0.912 – 16.7%</u>	0.693 – 0.842 – 15.8%
Qwen2.5-3B	0.447 – 0.711 – 40.4%	0.298 – 0.430 – 18.4%	0.509 – 0.754 – 28.1%	0.614 – 0.825 – 22.8%	0.561 – 0.772 – 22.8%
Llama-3.1-405B	0.772 – 0.860 – 9.6%	<u>0.781 – 0.886 – 10.5%</u>	0.684 – 0.789 – 10.5%	0.798 – 0.886 – 8.8%	0.781 – 0.833 – 5.3%
Llama-3.3-70B	<u>0.737 – 0.868 – 14.0%</u>	0.772 – 0.877 – 10.5%	0.632 – 0.702 – 7.0%	0.737 – 0.851 – 11.4%	<u>0.772 – 0.895 – 12.3%</u>
Llama-3.1-8B	0.667 – 0.807 – 14.9%	0.684 – 0.789 – 10.5%	0.658 – 0.825 – 16.7%	0.754 – 0.868 – 11.4%	0.746 – 0.842 – 9.6%
Llama-3.2-3B	0.553 – 0.825 – 28.9%	0.623 – 0.798 – 17.5%	0.386 – 0.544 – 15.8%	0.649 – 0.868 – 21.9%	0.737 – 0.860 – 12.3%

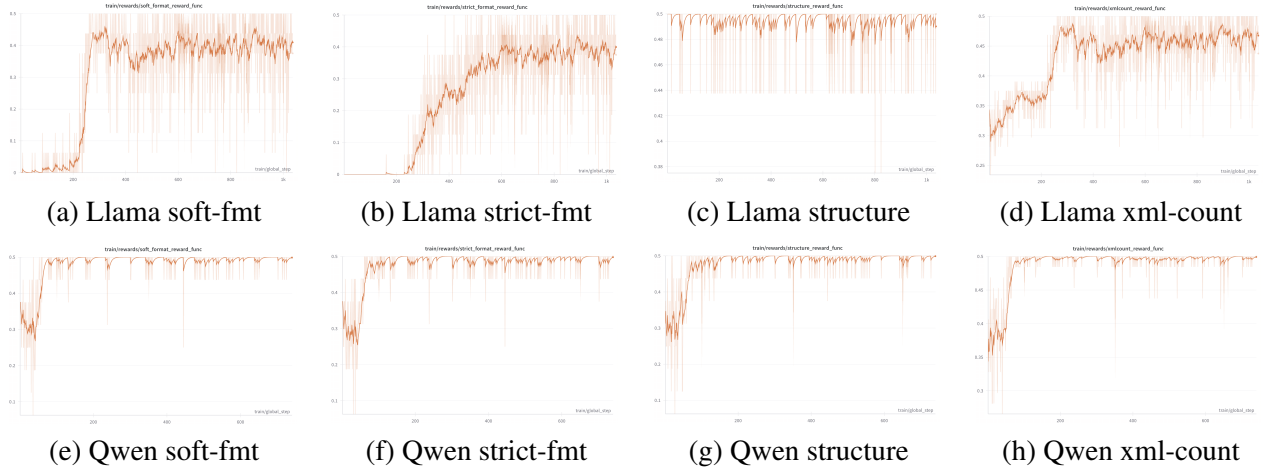


Figure 2: Evolution of the four auxiliary rewards (SOFT-FMT, STRICT-FMT, STRUCTURE, XML-COUNT) during GRPO fine-tuning. Shaded regions show batch-level variance; solid lines are exponential-moving averages. Maximum possible reward for each component = 0.5.

COUNT tracks the strict checker and plateaus below its maximum.

- **Qwen-2.5-3B**—Panels (e–h). All four format/structure rewards snap to the ceiling within the first ~ 120 steps, suggesting that smaller back-bones are easier to coerce into rigid templates. This fast saturation frees gradient budget for the harder CORRECTNESS signal.

Formatting constraints are therefore *easy* to encode, whereas the correctness component remains the bottleneck. Future work should freeze format rewards once saturated—or pre-train with a brief formatting-only phase—and allocate more RL steps or higher-rank LoRA to correctness, especially for larger models such as Llama-3.